

Thimble

Proposal - Google Summer of Code 2017

Personal Details

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Project Proposal

Project Title

Collaboration in Thimble (formerly Thimble and Remote Mentorship)

Overview

The audience of Thimble largely consists of teachers, educators and students. One of the most requested feature in thimble is collaboration.

Right now only a single user can work on a project.

The aim of the project is to allow multiple users to edit the same project at the same time.

Some of the other features that may be added in the future to improve collaboration include video/audio chat, real time audio translation and revision history.

There have been [instances](#) of people migrating to platforms other than Thimble because of the lack of Collaboration features. Through this project, I would like to understand what exactly teachers expect from a language learning environment and implement the same in Thimble. I'll try to finish as many features as I can during my Google Summer of Code coding period and work on the rest after that as people keep suggesting better ones.

I strongly believe that along with providing better performance to the current audience, this project will help magnify the audience of Thimble.

I've been working on open source for almost two years now. It is always great when to see your code being used and appreciated by people. Until now I have worked on [individual projects](#), in our [campus open source group](#) and a few [open source organizations](#).

Mozilla has been the pioneer in promoting open standards on the internet. I have learned immensely from the contributions I've done to Thimble in the past few weeks. Other than implementing my project, through this project I would like to understand how open source work goes on in bigger organizations like Mozilla and become a part of this community.

Previous contributions to Thimble and Bracket

- [#1856](#) (merged): Fixes the issue of popup window overflowing to the left side
- [#1809](#) (merged): Redirects the user to his projects page after he signs in from the home page.
- [#651](#) (open): Added Inline 3-D parameters editor Widget.
- [#1929](#)(open): Added buttons for toggling the value of auto-complete
- [#678](#)(open): Added logic to allow toggling of auto-complete

Contributions to y-js

- [#7](#) (merged): Fixes an incorrect function call in y-text's CodeMirror Collaborator.

Previous Projects Around Collaboration

- [Vroom.tk](#)
 - A Virtual reality teaching application made in Microsoft's 24 hour hackathon
 - Technologies used
 - WebRTC
 - Aframe
 - The application lets teachers broadcast videos to students in VR.
 - Steps to try
 - Open [vroom.tk](#)
 - Start a lecture.
 - Provide links to students to join.
- Chess Wars
 - Multiplayer chess playing platform.

- Uses socket programming to relay events.
- Prototype of Collaboration in Codemirror using y-js
 - Y-js is a collaboration framework on the lines of Togetherjs.
 - Recently they added Codemirror support to allow collaboration in Codemirror
 - I made a simple project to describe collaboration in Codemirror for multiple instances of Codemirror running in the same window.
 - The project can be found [here](#). A live demo can be found [here](#).

Schedule

Milestones

1. Realtime Viewing :
 - a. Invite users to look at the changes you make.
 - b. They can view the changes in real time. They can't edit the project.
2. Comments : Allow teachers/students to comment on each other's work.
This can be used as a tool to review students.
3. Realtime editing : Multiple users editing a project at the same time.
4. Levels of Access
 - a. Edit access :
 - i. Permission level : Highest
 - ii. People with edit access can edit a document
 - iii. Can be implemented in two ways
 1. Generating a link : Anyone with the link will have access to edit that page.
 2. Username method : The admin of the page can allow certain users to edit the doc.
 - iv. People with edit access can view, comment on and edit the page.
 - b. Comment access :
 - i. Permission level : Medium
 - ii. People with comment access can comment on a document but can't edit it.
 - c. View access :
 - i. Permission level : Low

- ii. People with view access can only view a document. This can be used to make a code snippet public, and at the same time not allow people to edit the same.
- 5. Areas of access :
 - a. Limiting the access of a user to particular files
 - b. This can be used for security reasons, allowing a collaborator to access only those files he needs to work on.
- 6. Chat :
 - a. Chat is useful features to interact while collaborative editing
 - b. Togetherjs already has a chat module.
 - c. Some of the extensions to chat could include :
 - i. Referring a line
 - ii. Tagging another user
 - iii. Smilies and Reactions

Research Milestones in the Future

1. Include Revision History
 - a. Another feature requested by teachers is providing revision history.
 - b. This will allow teachers to review a student's work more efficiently.
2. Add video/audio chat.
3. Realtime translation to remove the language barrier during communication.

Timeline

1. 3rd April - 4th May :
 - a. Fixing bugs in brackets and thimble.
 - b. Getting familiar with the code.
2. 4th May - 31st May :
 - a. Community Bonding.
 - b. Understanding the structure of Thimble.
 - c. Getting familiar with the code of togetherjs.
3. 1st June - 25th June - **Realtime Viewing**
 - a. Add the functionality to allow users to share their project with others and observe changes in real time.
 - b. This would involve creating endpoints similar to ones used for publishing projects and configuring Togetherjs to emit events from the host user to connected clients.

- c. Add a share button that generates the link for sharing the project.
- 4. 26th June - 10th July- **Comments**
 - a. Decide on the structure of adding comments.
 - i. Inline comments
 - ii. Comment on a selection
 - b. Add second permission level, Comments.
- 5. 10th July - 25th July - **Realtime Editing**
 - a. Implement two way communication among clients.
 - b. Decide on protocols to resolve conflicts.
Eg : If both users place cursor at the same place and start typing.
 - c. Add protocols to maintain consistency among clients and the server.
This would mark the completion of the first iteration of Collaboration features.
- 6. 26th July - 31st July - **Add Thimble specific features to Togetherjs chat**
 - a. Togetherjs already includes code that relays chat among clients.
 - b. Adding tagging feature
 - c. Allow users to refer to lines, files. Clicking on these would take the user to the given line/file.
- 7. 1st Aug - 15th Aug - **Allow admin to limit access to specific files**
 - a. Up to this point, a user will have a common permission (View/Edit/Comment) to access all files.
 - b. Add feature to allow users to provide different level of access to different files.
- 8. 16th Aug - Final Evaluation - **Final Testing**

The timeline might change depending on the library used for Collaboration. For Eg : Y-js already has Codemirror support, while Togetherjs already includes chat. More details can be found in the next section

Technical Requirements

According to the discussions going on Thimble's github repository, the expected libraries for collaboration include

- 1. [Togetherjs](#)
- 2. [Y-js](#)
- 3. Google drive api

TogetherJS and Y-js can't synchronize JavaScript application directly. To implement collaborative editing, Operational transformation can be used.

Y-js recently implemented [Codemirror support](#).

I made a simple [prototype](#) which is active [here](#) to handle collaboration in multiple Codemirror instances. Thimble runs multiple instances of Codemirror (One for each file). One of the concerns about keeping different instances in sync has been addressed in the prototype.

A chat feature can be implemented along the same lines. [Here](#) is an example on y-js's github page implementing the same.

On similar lines, Codemirror support can be added to Togetherjs. Togetherjs api allows relaying of events through their communication channel.

Effectively we need to relay insert and delete events of Codemirror to all the connected peers.

Benefits of Using Togetherjs

- Sister project of the Mozilla.
- No security issues in transmitting thimble events to Togetherjs server because of the same reason.
- Already has a chat feature implemented. Only need to improve it for Thimble.

Benefits of Using y-js

- Y-js recently introduced Codemirror support to allow collaboration.
- This is yet to be implemented in Togetherjs.

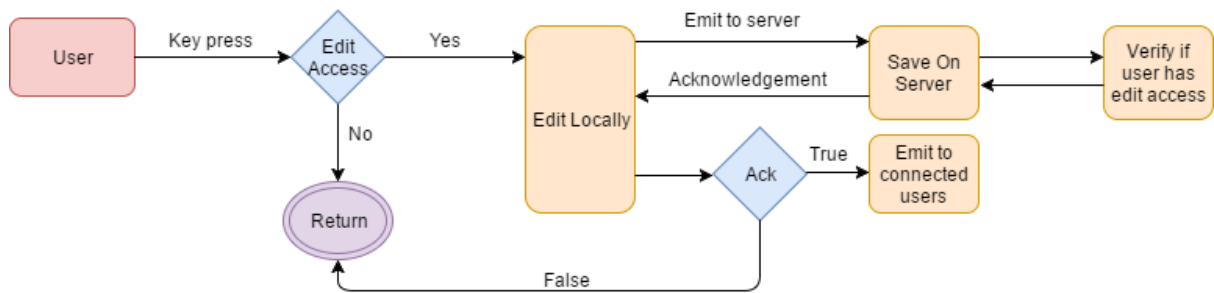
Whether to use Togetherjs, Y-js or rewrite something similar for Thimble is still under consideration. The final technology to be used will be decided after discussions with the mentors.

Engagements during Summer

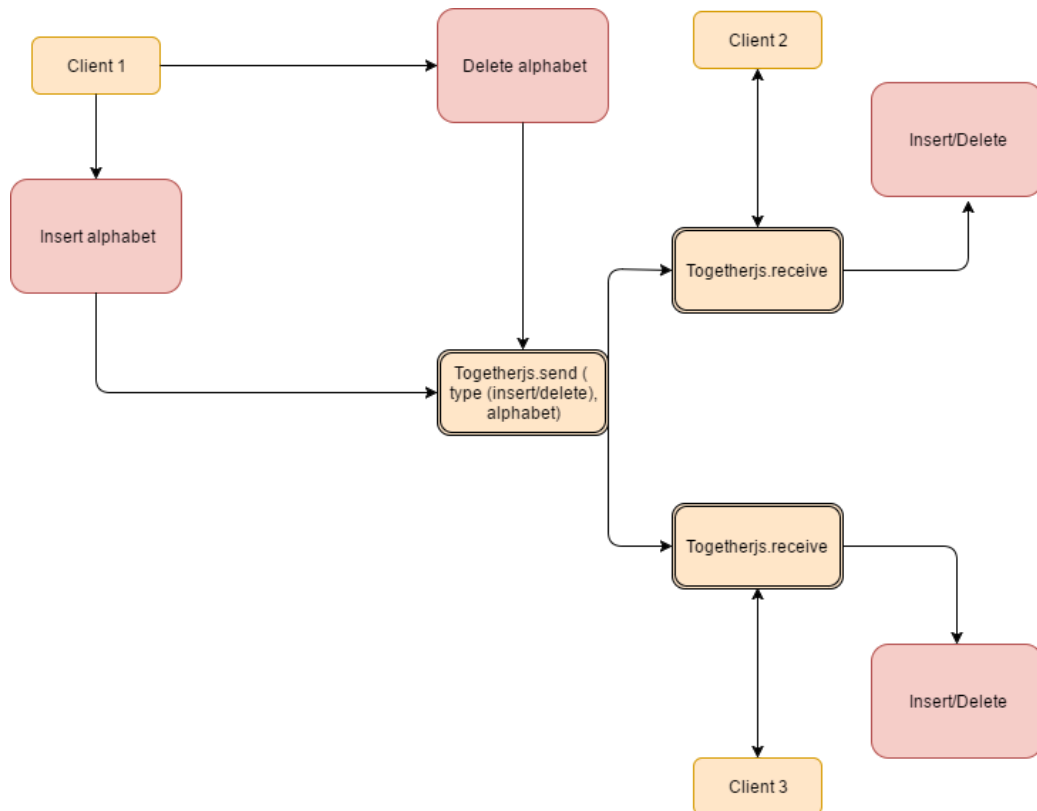
I have no commitments in the summer. I'll be staying back home for the most part of it. I have mentioned my typical working hours above and on an average will be able to spend 40 hours per week on the project.

Collaboration Flowchart

- Stage 1 - Realtime Viewing



- Stage 2 - Realtime Editing



Open Source Development Experience

I was introduced to open source two years back. Some of the open source projects that I have worked on include

1. [The Gambit Project, JSGTE](#) : I worked with the organization "The Gambit Project" during my Google Summer of Code last year. My work with the organization has been summarised [here](#).
All the work done is open source and is available on the github page of The Gambit Project.
2. [Notification Logger](#) : I built an open source tool useful for front end debugging. The project overrides the console.log functionality to provide custom logs as Desktop notifications.
It is available on my github page and has 900+ stars.
3. [P5.js](#) : I have been an active contributor to P5.js, an open source Javascript library for graphics. The library is a javascript extension of the famous graphics framework, Processing.
[Repository](#)
[My contributions](#)
4. Other than this, my other open source contributions can be found on my [github page](#).

Work/Internship/Academic Experience

- **Google Summer of Code 2016**

- Organization : The Gambit Project**

I participated in Google summer of code 2016. I worked with the organization [The Gambit Project](#). The [project](#) involved migrating the [Game Theory Explorer](#) from flash to Javascript. The details of the project and my contributions can be found [here](#). I am still a member of The Gambit Project, keep contributing to it and have regular meetups with the mentors.

- **Campus Activities**

I am pursuing B.Tech in Computer Science and Engineering from Indian Institute of Technology, Roorkee. I am currently in my pre-final year. I am the Joint Secretary of a campus group called [SDSLabs](#) where we develop software round the year for the people of IITR.

- Chess Wars : A chess playing platform live on the intranet of IIT Roorkee where people can play chess against their friends and against a bot. Front end is mainly javascript and CSS and the backend is written in Node. It involves the use of socket programming to handle asynchronous connections. The bot is written in python.
- Hubble : A link sharing platform for the people of IIT Roorkee. Live on the internet. Backend is written in PHP.